# KIIT UNIVERSITY

School of Computer Engineering

CAPSTONE PROJECT REPORT

# SAP SD Capstone Project

End-to-End Order-to-Cash (O2C) Process Implementation

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| **Module** | SAP Sales & Distribution (SD) |
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# 1. PROBLEM STATEMENT

Large supply chain and trading companies struggle to manage orders, inventory, and billing when their sales, warehouse, and finance teams operate on disconnected or manual systems. This results in:

- Delivery delays due to lack of real-time inventory visibility.
- Inaccurate invoices and billing errors from manual re-entry of data.
- Slow revenue collection caused by unlinked billing and payment processes.
- High operational costs from duplicate efforts and error correction cycles.
- Poor customer satisfaction from communication gaps between departments.

# 2. SOLUTION OVERVIEW

The solution is to implement SAP Sales and Distribution (SD) — the module that automates and connects the complete Order-to-Cash (O2C) pipeline. It provides a centralized, real-time platform where all teams share live, accurate data. To simplify understanding, the O2C flow maps directly to a modern Amazon-style e-commerce transaction:

| | | |
|---|---|---|
| 1 | **Customer Master (XD01)** | Creating your buyer account |
| 2 | **Inquiry (VA11)** | Checking product availability |
| 3 | **Quotation (VA21)** | Receiving a confirmed price & delivery date |
| 4 | **Sales Order (VA01)** | Clicking 'Place Order' |
| 5 | **Delivery & PGI (VL01N)** | Warehouse packs and dispatches the item |
| 6 | **Billing / Invoice (VF01)** | Invoice/receipt is generated and sent |
| 7 | **Incoming Payment (F-28)** | Payment is confirmed and account settled |

This capstone configures this exact flow for GlobalTech Innovations (Wholesale Hardware Division).

# 3. O2C PROCESS DIAGRAMS

## A. Complete O2C Flow

| **Order-to-Cash — End-to-End Flow** |
|---|

```
┌──────────────┐      ┌──────────────┐      ┌──────────────┐
│ Customer     │──────►│ Inquiry      │──────►│ Quotation    │──────►│ Sales Order
│ Master (XD01)│      │ (VA11)       │      │ (VA21)       │      │ (VA01)
└──────────────┘      └──────────────┘      └──────────────┘
                                                                    │
                                                                    ▼
                      ┌──────────────┐      ┌──────────────┐
                      │ Payment      │◄─────│ Billing      │◄─────│ Delivery & PGI
                      │ (F-28)       │      │ (VF01)       │      │ (VL01N)
                      └──────────────┘      └──────────────┘

Key: ──────► Document created with reference to previous step (zero re-
                                  entry)
```

## B. Cross-Module Integration

**SAP Module Integration Points**

```
                    ┌──────────────────┐
                    │     SAP SD       │
                    │ (Sales & Distribution│
                    │     Core Module) │
                    └──────────┬───────┘
                               │
                    ┌──────────┴──────────┐
                    ▼                     ▼
┌──────────────────────────┐   ┌──────────────────────────┐
│ SAP MM (Materials Mgmt.) │   │ SAP FI (Financial Acctg.) │
│ ATP Check at Sales Order │   │ Journal Entry on Billing  │
│ Stock Reduction on PGI   │   │ Open Item Cleared on Payment│
└──────────────────────────┘   └──────────────────────────┘
```

# 4. O2C PROCESS — STEP-BY-STEP EXECUTION

Each step below was executed in SAP and is documented with T-Code, process summary, execution sequence, and a screenshot placeholder.

## Step 1: Customer Master Creation [ T-Code: XD01 / BP ]

Creates the buyer profile in SAP — storing address, payment terms, and reconciliation account. Required before any transaction can begin.

**Execution Steps:**

1. Enter T-Code XD01 (or BP for Business Partner approach).
2. Select account role: Sold-to Party. Enter Sales Area and Company Code.
3. Complete General Data tab (name, address, contact).
4. Complete Company Code Data (reconciliation account — FI integration).
5. Complete Sales Area Data (payment terms, pricing, delivery priority).
6. Save. System generates a unique Customer Master Number.

## Step 2: Inquiry [ T-Code: VA11 ]

Customer's initial, non-binding request to check product availability and pricing. No financial commitment at this stage.

**Execution Steps:**

1. Enter T-Code VA11. Select Inquiry Type: IN.
2. Enter the Customer Number (from Step 1).
3. Enter Material Number and requested quantity.
4. Review auto-populated pricing data. Save.
5. System generates a unique Inquiry Document Number.

## Step 3: Quotation [ T-Code: VA21 ]

A legally binding offer specifying confirmed price, quantity, and validity date. Created with reference to the Inquiry — all data copies over automatically, eliminating re-entry.

**Execution Steps:**

1. Enter T-Code VA21. Select Quotation Type: QT.
2. Click 'Create with Reference' → enter Inquiry Number.
3. System copies all material, quantity, and pricing data automatically.
4. Set the Valid-to Date. Review and save.
5. System generates a unique Quotation Number.

## Step 4: Sales Order [ T-Code: VA01 ]

Official purchase confirmation — the central document of the O2C process. SAP performs an automatic ATP (Available-to-Promise) check to confirm inventory availability.

**Execution Steps:**

1. Enter T-Code VA01. Select Order Type: OR (Standard Order).
2. Click 'Create with Reference' → enter Quotation Number.
3. All data copies from Quotation to Sales Order.
4. SAP MM performs automatic ATP check — confirms stock availability.
5. System confirms delivery date. Save. Sales Order Number generated.

## Step 5: Outbound Delivery & Post Goods Issue [ T-Code: VL01N ]

Warehouse picks and ships the goods. Post Goods Issue (PGI) formally confirms the stock has left the facility — reducing inventory in SAP MM.

**Execution Steps:**

1. Enter T-Code VL01N. Enter Shipping Point and Goods Issue date.
2. System pulls material and quantity from the Sales Order.
3. Perform Picking: confirm the physically picked quantity.
4. Execute Post Goods Issue (PGI) — stock reduced in SAP MM.
5. Goods Movement document created automatically for audit trail.

## Step 6: Billing / Invoice [ T-Code: VF01 ]

Generates the invoice after PGI is complete. On saving, SAP FI automatically posts a journal entry: Debit Customer A/R, Credit Revenue Account.

**Execution Steps:**

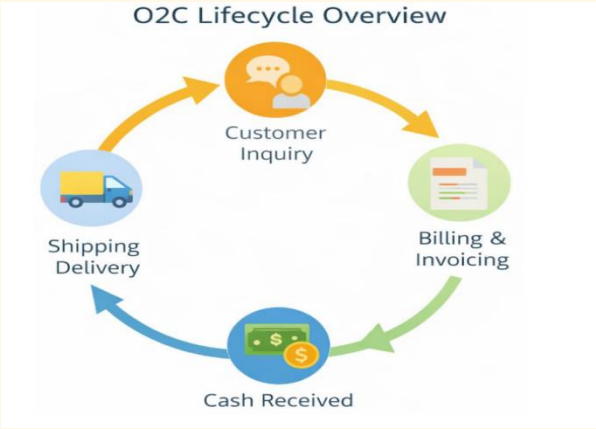
1. Enter T-Code VF01. System auto-retrieves the Delivery document.
2. Verify pre-populated material, quantity, and pricing data.
3. System calculates applicable taxes and splits revenue accounts.
4. Save. Billing Document Number generated.
5. SAP FI posts automatic journal entry (Dr. Customer A/R / Cr. Revenue).

### Step 7: Incoming Payment [ T-Code: F-28 ]

Finance team records the customer's payment, clears the open invoice, and completes the O2C cycle. After this step, the customer's account balance is zero.

**Execution Steps:**

1. Enter T-Code F-28. Enter Bank Account details and amount received.
2. Enter posting date and value date of the payment.
3. In Open Items, select the outstanding invoice to clear.
4. Verify amount matches invoice — difference must be zero.
5. Save. Customer account cleared. O2C cycle complete.




## 5. TECHNOLOGY STACK

| | |
|---|---|
| **Platform** | SAP S/4HANA (compatible with SAP ECC 6.0) |

| | |
|---|---|
| **Core Module** | SAP SD — Sales and Distribution |
| **Integration: MM** | SAP Materials Management — ATP check & stock reduction on PGI |
| **Integration: FI** | SAP Financial Accounting — auto journal entries & payment clearing |
| **Config Tool** | SAP SPRO (Implementation Guide) — module customization |
| **Interface** | SAP GUI / SAP Fiori (future roadmap) |

# 6. UNIQUE POINTS

## 6.1 Real-World Conceptual Mapping

Every step of the O2C process is mapped to an Amazon-style e-commerce transaction. This makes the entire ERP workflow immediately understandable to any audience — technical or non-technical — and demonstrates strong business thinking beyond just system configuration.

## 6.2 Zero Data Redundancy — Document Flow

The 'Create with Reference' chain (Inquiry → Quotation → Sales Order) means data is entered exactly once and carried forward automatically. This eliminates re-entry errors, speeds up transaction processing, and is a hallmark of professional SAP configuration that distinguishes this project from basic implementations.

## 6.3 Cross-Functional Module Integration

This project explicitly demonstrates two live SAP integration touchpoints: (1) SAP MM is triggered at Delivery (PGI auto-reduces stock), and (2) SAP FI is triggered at Billing (automatic journal entry posted). Proving cross-module integration is what separates an O2C project from a single-module exercise.

## 6.4 AI-Based Demand Forecasting (Advanced Concept)

As a forward-looking enhancement, integrating SAP Business AI or an external ML model with the SD order history data would enable predictive demand forecasting — automatically recommending reorder quantities, alerting on potential stock-outs before they occur, and optimizing pricing based on seasonal trends. This positions the project at the intersection of ERP and intelligent automation.

## 6.5 Exam Memorability — The C-I-Q-S-D-B-P Framework

The seven steps are encoded as the acronym C-I-Q-S-D-B-P (Customer, Inquiry, Quotation, Sales Order, Delivery, Billing, Payment). This structured recall framework enables confident, error-free presentation of the full O2C cycle under examination conditions.

## 7. FUTURE IMPROVEMENTS

- SAP Fiori Integration: Migrate key transactions (VA01, VL01N, VF01) to Fiori apps for a role-based, mobile-friendly interface for field sales and warehouse teams.
- API Automation via OData: Expose SD transactions as RESTful OData services to allow e-commerce platforms and CRM systems to trigger the O2C cycle without manual SAP entry.

## 8. GITHUB REPOSITORY

**Repository URL: https://github.com/Purushottam0001/sap-sd-o2c-**

Visibility: Public | Language: SAP Configuration / Documentation

The repository contains the following:

- config/ — Step-by-step SAP configuration notes (Sales Area, Customer Master, Pricing Conditions, Output Types).
- screenshots/ — Full execution proof: one folder per T-Code (XD01, VA11, VA21, VA01, VL01N, VF01, F-28).
- docs/ — This project report in PDF and DOCX format.
- diagrams/ — O2C flow diagram and module integration map (exported as PNG).
- README.md — Project overview, business scenario (GlobalTech Innovations), and setup instructions.